

- Q.31 How the level measurement by using capacitive transducer.
- Q.32 Explain the method of level measurement by using ultrasonic method.
- Q.33 Describe any one method for low pressure measurement.
- Q.34 Explain the working principle of turbine flow meter.
- Q.35 Describe the diaphragm method for pressure measurement with its applications.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain construction, principle and working of thermocouple.
- Q.37 Write a short notes in
 a) Bellows b) diaphragm
 c) Burdon Tube
- Q.38 Describe construction and working principle of venturimeter and orifice for flow Measurement.

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5th Sem / IC, EI Subject:- Process Instrumentation

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 High pressure measurement by
 a) Mano meter b) Burdon tube
 c) Pirani gauge d) Bridgman gauge
- Q.2 Unit of Capacitor
 a) Fard b) ohm
 c) Henry d) None of these
- Q.3 Hot wire anemometer is used for
 a) Temperature b) flow
 c) Pressure d) Level
- Q.4 Meaning of Pyro is _____
 a) Light b) Wight
 c) Liquid d) Radiation
- Q.5 Rotameter is used to measure _____
 a) Level b) Temperature
 c) Flow d) Difference

- Q.6 Working Principle of thermocouple is _____
- Halls Effect
 - Electromagnetic Radiation
 - Seeback effect
 - None of these
- Q.7 Unit of frequency is _____
- Hz
 - Meter
 - Second
 - Pascal
- Q.8 Which of the following temperature sensor used for high range application? _____
- RTD
 - thermocouple
 - thermistor
 - gas thermometer
- Q.9 Low pressure measurement by _____
- Bellows
 - Pirani gauge
 - ionization gauge
 - both b & c
- Q.10 Level measurement by _____
- Resistive transducer
 - inductive transducer
 - Capacitive transducer
 - All of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Thermo couple is _____ transducer.
- Q.12 Write two unit of pressure.
- Q.13 Principle of thermocouple based is.
- Q.14 U-tube manometer is used for _____

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- Q.15 Bridgman gauge is use to measure _____ pressure.
- Q.16 Rota meter is always placed in vertical position (Yes/No)
- Q.17 Orifice is used to measure _____ rate.
- Q.18 What is inductance.
- Q.19 Principle of pirani gauge is based on
- Q.20 What is the unit of frequency.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the working principle of U tube manometer.
- Q.22 Discuss the fiber optics thermometer in detail.
- Q.23 Explain the McLeod gauge drive the expression of pressure.
- Q.24 Draw the diagram of inclined type and well type manometer.
- Q.25 Discuss the Hot wire anemometer in detail.
- Q.26 Explain working principle of radiation pyrometer.
- Q.27 Discuss the working principle of Rotameter and write the force balance equation.
- Q.28 Describe the Rotameter method for flow measurement.
- Q.29 Write short note on Electromagnetic flow meter.
- Q.30 What is pressure, write the relation between pressure and force and also write four units of pressure.

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